Last updated: April 2026

Git Documentation & Organization

Goal: The goal of the document is to provide information on the files available in Github, as well as those we could not make publicly available, so that a user could understand what information they would need to compile to run the model or apply it elsewhere.

Data Availability: This repository includes all of the novel code and public data developed for this work. However, there are two proprietary third-party datasets from the City of Santa Cruz that are available to the authors under a data use agreement but cannot be made public due to privacy and security concerns: household water billing data and some details (e.g., nodes and edges) from the City's water supply model.

Notes: We detail below which files are not able to be publicly shared, and we detail the necessary data columns or other relevant information to recreate them.

Subfolder: data

costs

- Name: Cost_SCWSM_LTVA.csv
    - Purpose: Csv file with cost values for deployment order in linear programming

data_system

- Name: Stage_Storage_Area_MG_2020_bathymetry.csv
    - Purpose: Csv file with bathymetry information for Loch Lomond

dcc_data

- Name: DCCcoefs.csv
    - Purpose: Csv file with DCC model coefficients
- Name: household_income_data.csv
    - Purpose: Csv file with single-family households data for use in simulation model
    - Columns: account, pipe size (in), pool, bathrooms, main area (sqft), tax value, census tract/block group, map_inc_1 … map_inc_10, mean residuals (real)
    - Note: file not included in Github repository

Input_climate_stress_test.zip

- Note: This data is too large for Github and so it is directly linked to Zenodo.
- Subfolder: FLOW
    - Purpose: Csv files with the streamflow data for each stochastic simulation

            - Columns: Date; Streams (raw): Bigtrees, Laguna, Liddel, Majors, Newell, Taitside; Streams (Na): Bigtrees, Laguna, Liddel, Majors, Taitside; is it a critically dry year? (binary); first flush (binary)
- Subfolder: WEATHER
    - Purpose: Csv files with the weather data for each stochastic simulation
        - Columns: Date, Precipitation, Evaporation, Turbidity (San Lorenzo River), Turbidity (North Coast), Turbidity (North Coast Majors)
            - Note: Turbidity is binary

climate_monthly.zip

- Contains: files for each stochastic climate scenario with monthly precipitation, temperature, and evapotranspiration data for us in the econometric model

Subfolder: model_assumptions_and_scenarios

- Name: cashflow_rate_assumptons.json
    - Purpose: JSON file with assumptions related to the cashflow model
- Name: inf_planning_assumptions.json
    - Purpose: JSON file with assumptions related to the infrastructure financing model, including infrastructure options (costs, deployment time, etc.)
- Name: model_assumptions_CSTE.csv
    - Purpose: Csv file with water resources model assumptions involving single values
    - Columns: Assumption (name of assumption), SCWSM-Option_Analysis… (version name, most are based on the demand scenario, examples include: SCWSM-Demand_Baseline)
    - Note: contains proprietary data not able to be included in Github repository
- Name: model_assumptions_PROFILES.xlsx
    - Purpose: Excel file with water resources model assumptions requiring monthly profiles
    - Columns: Assumption (name); 1, 2, … (month of year)
    - Tabs: Names for each version, i.e., SCWSM-Option_Analysis
    - Note: contains proprietary data not able to be included in Github repository
- Name: used_assumptions_CSTE.csv:
    - Purpose: contains the single values utilized in the model
    - Note: File created in simulation setup
- Name: used_assumptions_PROFILES.csv
    - Purpose: contains the monthly profiles utilized in the model
    - Note: File created in simulation setup
- Name: Hydrological_Condition_Types.csv
    - Purpose: Csv file containing the values for the hydrological condition type by month of year
- Subfolder: sensitivity_analysis

    - Purpose: contains all of the json files to set up the sensitivity analysis parameters and simulations. Naming convention for file names is as follows:
        - Cashflow_rate… files include relevant assumptions for the cashflow model in different analyses
        - Inf_planning… files contain relevant assumptions for the infrastructure financing model
        - SA_… files contain the relevant file names and parameters for different sensitivity analyses
    - Naming conventions for sensitivity analyses is as follows:
        - Baseline: baseline simulations
        - CoS: High/low Cost-of-service sensitivity analysis
        - Demands: High/low residential demands sensitivity analysis
        - DealTime: Short/long desalination deployment time sensitivity analysis
        - InfCosts: High/low infrastructure costs sensitivity analysis
        - InterestRate: High/low interest rate sensitivity analysis

Subfolder: model > SCWSM-Option_Analysis > Parent

- Name: CST_SCWSM_OPTION_ANALYSIS.json
    - Purpose: JSON file containing the parameters used to build the Pywr model for the baseline simulations
    - Notes: Parts of this file have been removed including all nodes, all edges, some parameters, and some recorders
- Name: CST_SCWSM_OPTION_ANALYSIS_SA.json
    - Purpose: JSON file containing the parameters used to build the Pywr model for the sensitivity analysis simulations
    - Notes: Parts of this file have been removed including all nodes, all edges, some parameters, and some recorders
- Note: create subfolder to save individual json files into for stochastic simulations

Subfolder: scripts

- Name: Setup_SCWSM_Option_Analysis_CST.py
    - Purpose: Python file that sets up the Pywr model for that specific simulation, including importing the climate data and the sensitivity analysis parameters
- Name: household_demands.py
    - Purpose: Python file that defines the class householdDemand
- Name: scwsm_demands.py
    - Purpose: Python file that contains all of the parameter classes input into Pywr built for this project and paper
- Name: scwsm_alternative_supply_option_operations.py
    - Purpose: Python file with the parameter classes for the alternative water supply options and operations
    - Parameter classes for specific water resources system removed, including:

        - SUPPLY_GAP: specific to SCWD
        - FELTON_DIV_TO_LL: specific to SCWD
        - DESALINATION_PLANT_YIELD/DESALINATION_PLANT_YIELD_ROF,: determines the desalination plant yield based on the month of the year and storage levels
        - DIRECT_POTABLE_REUSE_PLANT_YIELD: determines the DPR plant yield based on the month of the year and storage levels
        - DEMAND_AFTER_RATIONING: determines the value of reduced urban demand depending on if rationing/curtailment is implemented in the model
        - TRANSFER_YIELD: returns the transfer volume from x neighboring water district based on reservoir levels
        - MCGWB_SWITCH: a switch that turns the capacity of extraction to 0 on May 1st based on reservoir storage
        - Soquel_SWITCH: a switch that turns the supply from Soquel ASR on/off based on reservoir levels- checked between May and September
- Name: scwsm_model_assumptions.py
    - Purpose: Python file that imports assumptions from relevant csv files
    - Note: Csv files imported not included in Github repository
- Name: scwsm_operations_and_constraints.py
    - Purpose: Python file that includes parameter classes for relevant operations and constraints
    - Parameter classes:
        - Max_flow_below_LL: sets the maximum flows downstream of the reservoir based on the month of the year
        - FELTON_DIV_TO_LL: calculates the flow pumped from Felton to the reservoir
        - Hydraulic_constraint: calculates the hydraulic loss factor for a pump in the system
    - Note: not included in Github repository
- Name: scwsm_water_rights.py
    - Purpose: Python file that includes parameter classes for relevant water rights assumptions and operations
    - Parameter classes:
        - Felton_to_LL_Annual_License: calculates the annual license at a diversion within the system
        - WATER_RIGHT_NEWELL_TO_GHWTP: returns current value of reservoir license tied to use of upstream source for water supply
        - WATER_RIGHT_NEWELL_DIV_TO_LL: returns current value of license tied to certain diversion
        - PRECIP_WATER_IN_LL: returns the volume of water from local rainfall over the reservoir
        - FELTON_WATER_IN_LL: returns the volume of water in the reservoir originating from a certain source

        - NEWELL_WATER_IN_LL: calculates the volume of water in the reservoir from a specific upstream source
        - USEABLE_PRCP_TO_GHWTP: reuters volume available from precipitation in reservoir available for water treatment plant
        - WATER_MONTH_TYPE: returns the water month type based on flow from an uptream source
        - SWITCH_MCASR_INJECTION: returns the binary value based on the water month type parameter
        - SWITCH_TRANSFERS returns the binary value based on the water month type parameter
    - Note: not included in Github repository
- Name: run_simulations_caps.py
    - Purpose: Python file that runs the simulations used in the paper
    - Note: run on HPC system; you can run it on a local machine, but can comment lines to run different sets of policy simulations. On a local machine, it will take 1-2 minutes per simulation.
- Subfolder: plots
    - Name: Figure1_ModelSchematic.ipynb
        - Purpose: Python file to create paper plot Figure 1 using two schematics developed in PowerPoint
    - Name: Figure2_assistance_dynamics.py
        - Purpose: Python file to process the data for and create paper plot Figure 2
    - Name: Figure3_compare_climate_scenarios.py
        - Purpose: Python file to create paper plot Figure 3
    - Name: Figure4_cost_drivers.py
        - Purpose: Python file to process and plot Figure 4
    - Name: Figure5_households_with_assistance.py
        - Purpose: Python file to process and plot Figure 5
    - Name: Figure6_Sherlock.py
        - Purpose: Python file to process data for and plot Figure 6
        - Note: This figure was made on an HPC computing system
    - Name: Figure7_scattter.py
        - Purpose: Python file to process data for and plot Figure 7
    - Name: processing_functions.py
        - Purpose: Python file with functions for processing the different types of data across results figure files
    - Name: SI_Boxplots_ParameterJustification.py
        - Purpose: Create Figure S2, which uses boxplots to justify the assistance cost parameters used
    - Name: SI_StochasticSimulations.py
        - Purpose: Create Figure S3, showing sample stochastic climate simulations
    - Name: SI_SensitivityAnalysis.py

        - Purpose: Creates Figure S4, which compares distributions of various metrics across the different sensitivity analyses and then computes the statistics included in Table S7
    - Name: SI_Figure5_Hypothesis.py
        - Purpose: Create Figure S5, which compares the annual assistance cost and number of infrastructure options deployed across moderate and dry climate scenarios
    - Name: SI_Table_Cost_Data.py
        - Purpose: obtain the data for Table S8
    - Name: SI_SensitivityAnalysis_IE_Table.py
        - Purpose: process and output the data for Table S9
    - Name: Figure3_IE_Sherlock.py
        - Purpose: create an alternative version of Figure 3 with a different income estimate for the SI
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        - Purpose: create an alternative version of Figure 5 with a different income estimate for the SI
    - Name: Figure6_IE_Sherlock.py
        - Purpose: create an alternative version of Figure 6 with a different income estimate for the SI
    - Name: Figure7_IE_Sherlock.py
        - Purpose: create an alternative version of Figure 7 with a different income estimate for the SI

Subfolder: results

- Name: df_cashflow_Baseline_P100T0_dCV1.0_real1270_demandBaseline.csv
    - Purpose: Csv file containing monthly cashflow model simulation results
- Name: df_monthly_assistance_P100T0_dCV1.0_real1270_demandBaseline.csv
    - Purpose: Csv file containing the amount of monthly assistance by assistance type
- Name: df_results_Baseline_P100T0_dCV1.0_real1270_demandBaseline.csv
    - Purpose: Csv file containing daily water system simulations results
- Name: df_time_tracker_Baseline_P100T0_dCV1.0_real1270_demandBaseline.csv
    - Purpose: Csv file containing the infrastructure options deployed, their costs, and timing.

- Note: These are sample water system results for a single realization are included in the repository. We have only included results files for data that is not aggregated at the household level.